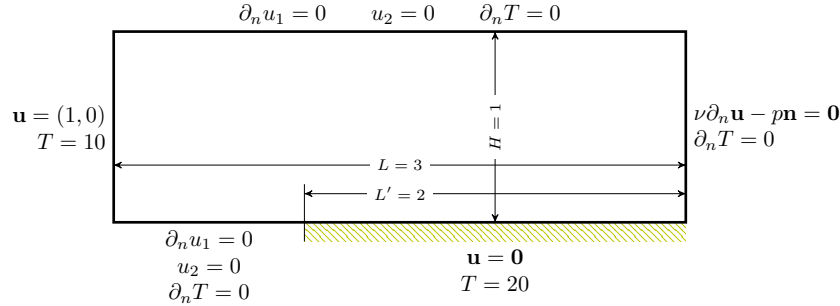


CFD 2025-26 – Lab 11

Exercise 1

We want to simulate the thermal boundary layer near a hot plate. The flow is governed by the following steady incompressible Navier–Stokes equations with the *Boussinesq buoyancy model* in the domain represented here, together with the boundary conditions:



$$\begin{cases} -\nu \Delta \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p + \beta \mathbf{g}(T - T_{\text{ref}}) = 0, & \text{in } \Omega, \\ \text{div } \mathbf{u} = 0, & \text{in } \Omega, \\ -k \Delta T + \mathbf{u} \cdot \nabla T = 0, & \text{in } \Omega, \end{cases} \quad (1)$$

where $\mathbf{u}$, p and T are velocity, pressure and temperature, respectively, ν is the kinematic viscosity, β is the thermal expansion coefficient, $\mathbf{g} = (0, -9.8)$ is the gravity acceleration, T_{ref} is a reference temperature and k the thermal diffusion coefficient.

The boundary conditions represent:

- inlet: a flow of fluid at $T = 10$ entering the domain with a flat velocity profile;
- outlet: “no-stress” conditions on the flow and no-flux (adiabatic) conditions on temperature;
- a rough (no-slip) solid plate at $T = 20$;
- symmetry conditions on both velocity and temperature on the top boundary and on the part of the bottom boundary preceding the plate (i.e. the plate is suspended).

- 1 Define a fixed-point iteration method to solve system (??), separating the flow problem from the thermal one and properly linearizing them. Write the weak formulation of the differential problems to be solved at each iteration and subsequently their finite element approximation.
- 2 Solve the problem in Firedrake, using the stable pair of spaces $\mathbb{P}_2/\mathbb{P}_1$ for the discretization of stationary Navier-Stokes and the space $\mathbb{P}_1$ for the temperature field, with a spatial discretization parameter given by $h = 0.1$. The other parameters are given by

$$\nu = 0.1, \quad k = 10^{-2}, \quad \beta = 10^{-3}, \quad T_{\text{ref}} = 0.$$

Consider the initial guess $\mathbf{u}_0 = \mathbf{i}, p_0 = 0, T_0 = 10$ and the following stopping criterion based on the relative increment

$$\frac{\|\mathbf{u}_k - \mathbf{u}_{k-1}\|_{H^1}}{\|\mathbf{u}_{k-1}\|_{H^1}} + \frac{\|T_k - T_{k-1}\|_{H^1}}{\|T_{k-1}\|_{H^1}} \leq 10^{-4}.$$

- 3 Fill the following table for different values of the thermal diffusivity.

k	10 ⁻²	10 ⁻³	10 ⁻⁴
Fixed-point iterations			

What can you observe in the temperature distribution?

- 4 Introduce a proper stabilization where necessary and give a description of the stabilization technique introduced. Then, repeat the previous point.

k	10 ⁻²	10 ⁻³	10 ⁻⁴
Fixed-point iterations			